

New York City Taxi Rides

Power BI Business Intelligence Project

Course: DSA3050A – Business Intelligence & Data Visualization **Students:**

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Week 6 - Proposal

Problem Statement

New York City's taxi operations face challenges in optimizing fleet utilization, understanding passenger demand patterns, and maximizing revenue across different neighborhoods and time periods. This project analyzes NYC taxi trip data to identify operational inefficiencies, peak demand periods, and neighborhood-level performance metrics. The goal is to provide data-driven recommendations for improving taxi availability, reducing passenger wait times, and increasing driver revenue through better route and timing decisions.

Proof of Dataset legitimacy

-Taxi rides dataset url(Taxi)

 [kaggle.com/datasets/surekharamireddy/new-york-city-taxi-rides](https://www.kaggle.com/datasets/surekharamireddy/new-york-city-taxi-rides)



- Taxi rides website evidence

New York City taxi rides

Develop a predictive model to forecast taxi ride duration.



Data Card

Code (0)

Discussion (0)

Suggestions (0)

About Dataset

Description

New York City taxi rides form the core of the traffic in the city of New York. The many rides taken every day by New Yorkers in the busy city can give us a great idea of traffic times, road blockages, and so on. A typical taxi company faces a common problem of efficiently assigning the cabs to passengers so that the service is hassle-free. One of the main issues is predicting the duration of the current ride so it can predict when the cab will be free for the next trip. Here the data set contains various information regarding the taxi trips, its duration in New York City.

Usability ⓘ

8.82

License

MIT

Expected update frequency

Never

Tags

- Data Visualization
- Data Cleaning

Business Questions

1. What are the peak hours and days for taxi demand across different neighborhoods?
2. Which neighborhoods generate the highest trip volumes and revenue?
3. How does trip distance correlate with fare amount and payment type?
4. What is the average trip duration and how does it vary by time of day?
5. Which payment methods are most common and how do they vary by neighborhood?
6. How many passengers typically ride together and does this affect trip distance?
7. What is the relationship between pickup and dropoff neighborhoods (commuter patterns)?

Entity Relationship Diagram Draft

Source Tables

trips_1.csv (Fact Table - 50,000+ rows)

- id (Primary Key)
- vendor_id
- pickup_datetime
- dropoff_datetime
- passenger_count
- trip_distance
- pickup_longitude

- pickup_latitude
- dropoff_longitude
- dropoff_latitude
- payment_type (1-6 code)
- trip_duration
- pickup_neighborhood (FK to pickup_neighborhoods)
- dropoff_neighborhood (FK to dropoff_neighborhoods)

pickup_neighborhoods.csv (Dimension)

- neighborhood_id (Primary Key)
- neighborhood_name

dropoff_neighborhoods.csv (Dimension)

- neighborhood_id (Primary Key)
- neighborhood_name

Planned Star Schema

Fact Table:

- FactTrips (id, date_key, vendor_id, passenger_count, trip_distance, payment_type, trip_duration, pickup_neighborhood_key, dropoff_neighborhood_key)

Dimension Tables:

- DimDate (date_key, date, year, month, day, day_of_week, hour, quarter)
- DimPickupNeighborhood (neighborhood_key, neighborhood_id, neighborhood_name)
- DimDropoffNeighborhood (neighborhood_key, neighborhood_id, neighborhood_name)
- DimPayment (payment_key, payment_code, payment_description)
- DimVendor (vendor_key, vendor_id, vendor_name)

Data Source

Dataset: New York City taxi rides **Source:** Kaggle (<https://www.kaggle.com/datasets/surekharamireddy/new-york-city-taxi-rides>) **Files:**

- trips_1.csv (Main trip records)
- pickup_neighborhoods.csv (Pickup location lookup)
- dropoff_neighborhoods.csv (Dropoff location lookup)

Data Citation

Ramireddy, S. (2023). New York City taxi rides [Data set]. Kaggle.
<https://www.kaggle.com/datasets/surekharamireddy/new-york-city-taxi-rides>

Part 1: Transformations and DAX

Week 8 - Power query Transformations

Loading data into POWER BI

loading all 3 datasets to powerBI for Transformation



dropoff_neighborhoods.csv

File Origin

1252: Western European (Windows) ▾

Delimiter

Comma ▾

Data Type Detection

Based on first 200 rows ▾

neighborhood_id	latitude	longitude
C	40.78378035	-73.95314487
S	40.83679153	-73.93438102
AL	40.81844494	-73.94804587
J	40.79957292	-73.94042746
AV	40.77627046	-73.98232156
D	40.75017915	-73.99255731
T	40.6987723	-73.81513781
U	40.79359677	-73.96982243
I	40.78500471	-73.97605049
AO	40.74712554	-73.98533557
AC	40.72024502	-73.98720493
AT	40.72966953	-73.98169332
Y	40.74253132	-73.97794274
F	40.77174819	-73.86895975
E	40.69748393	-73.92248289
AB	40.73059215	-74.00311431
K	40.76680909	-73.95688626
AS	40.69200157	-73.99131297

Extract Table Using Examples

Load

Transform Data

Cancel



pickup_neighborhoods.csv

File Origin

1252: Western European (Windows) ▾

Delimiter

Comma ▾

Data Type Detection

Based on first 200 rows ▾

neighborhood_id	latitude	longitude
AH	40.80434902	-73.96171613
Z	40.71582831	-73.95429804
D	40.75017915	-73.99255731
AT	40.72966953	-73.98169332
AG	40.74984348	-74.00345827
K	40.76680909	-73.95688626
R	40.73821114	-73.98754432
A	40.76149153	-73.97589934
AK	40.76472344	-73.96669585
U	40.79359677	-73.96982243
AB	40.73059215	-74.00311431
Q	40.74033267	-74.00510672

Extract Table Using Examples

Load

Transform Data

Cancel

pickup_neighborhoods.csv

File Origin

1252: Western European (Windows)

Delimiter

Comma

Data Type Detection

Based on first 200 rows

neighborhood_id

latitude

longitude

AH	40.80434902	-73.96171613
Z	40.71582831	-73.95429804
D	40.75017915	-73.99255731
AT	40.72966953	-73.98169332
AG	40.74984348	-74.00345827
K	40.76680909	-73.95688626
R	40.73821114	-73.98754432
A	40.76149153	-73.97589934
AK	40.76472344	-73.96669585
U	40.79359677	-73.96982243
AB	40.73059215	-74.00311431
Q	40.74033267	-74.00510672

Extract Table Using Examples

Load

Transform Data

Cancel

Renaming the Queries

- trips_1 → FACT_trips
- pickup_neighborhoods → DIM_Pickup_hood
- dropoff_neighborhoods → DIM_Dropoff_hood

 DIM_dropoff_hoods

 DIM_pickup_hoods

 FACT_trips

Transformations:

Promoting headers

On FACT_trips table

Original

Transformation

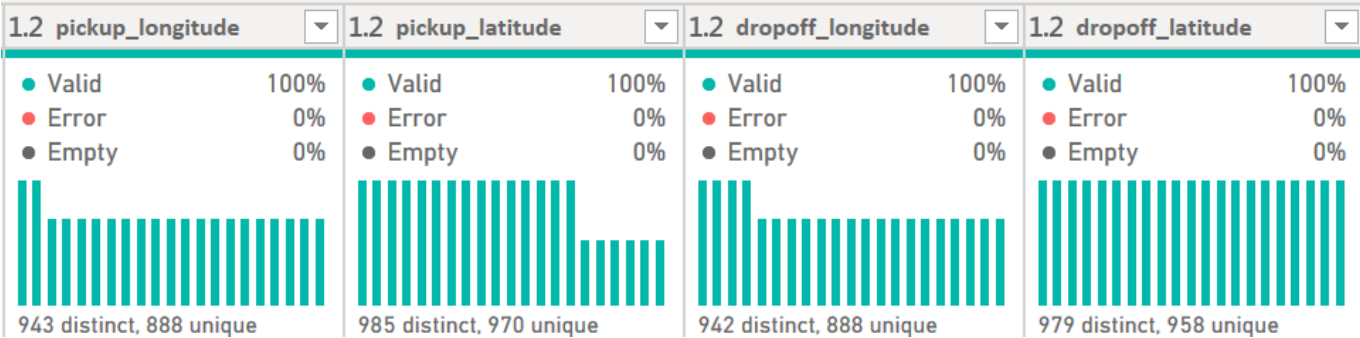
Final

Removing Unnecessary Columns

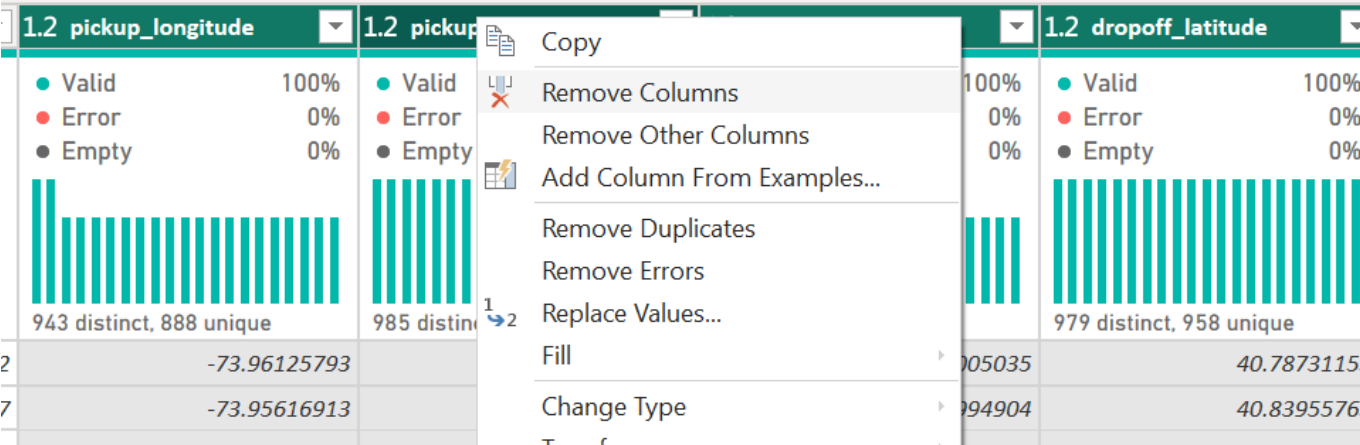
On FACT_trips table

We decided to get rid of pickup longitude and latitude as well as drop off longitude and latitude as they exist in the Dimension tables and are redundant for the FACT table.

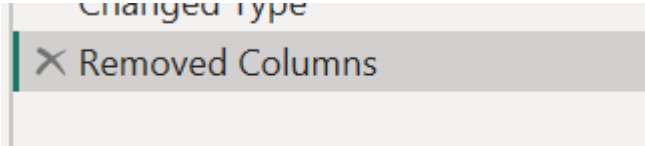
Original



Transformation



Final

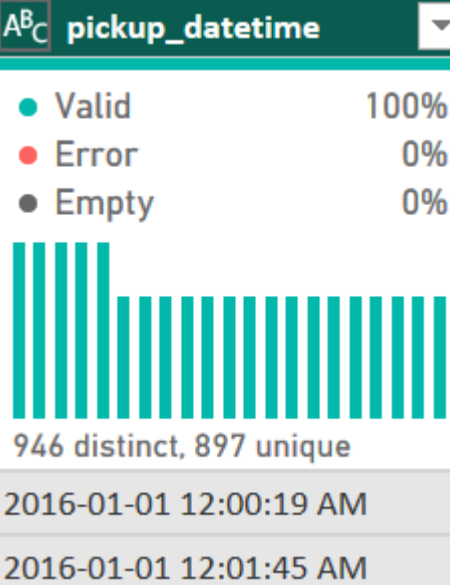


Fixing Data Types

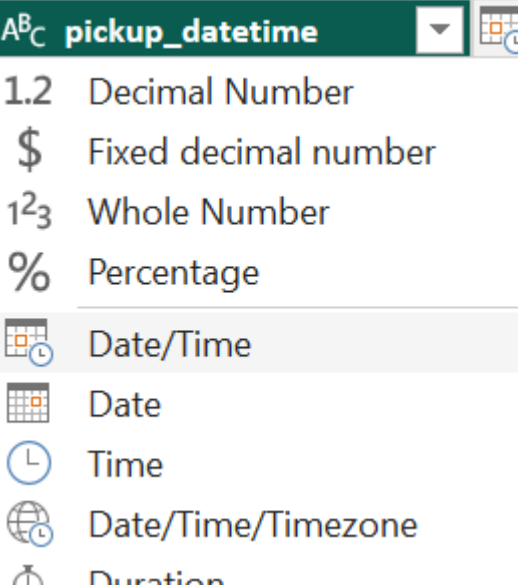
On FACT_trips table

Pickup Datetime

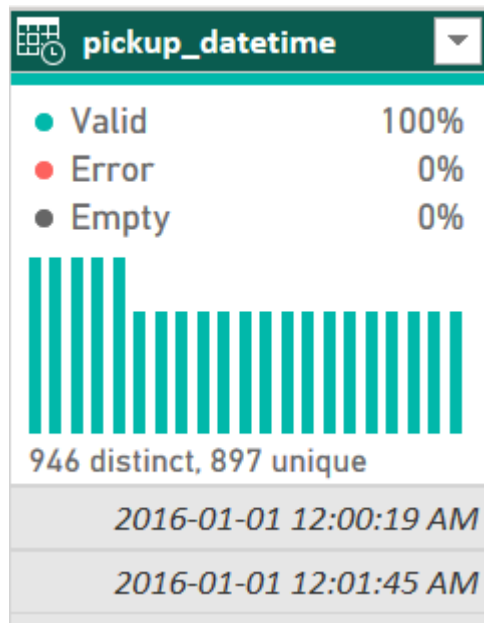
Original



Transformation

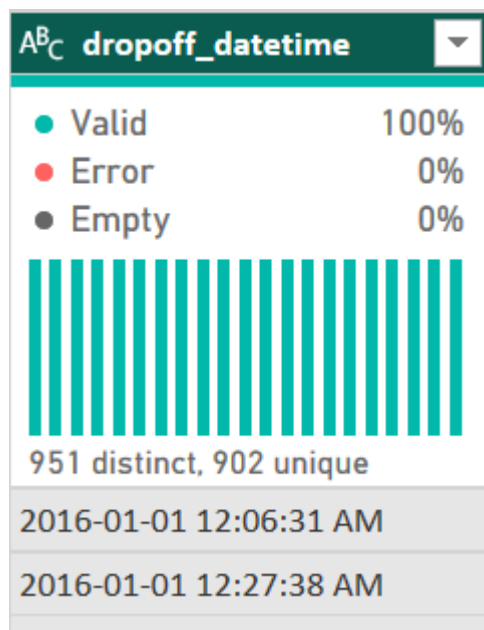


Final

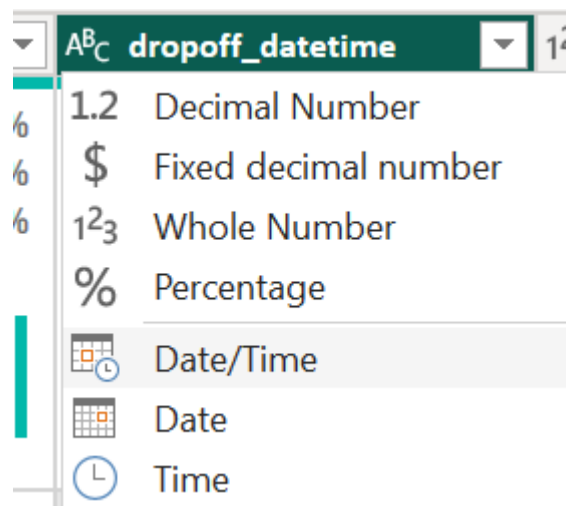


Dropoff Datetime

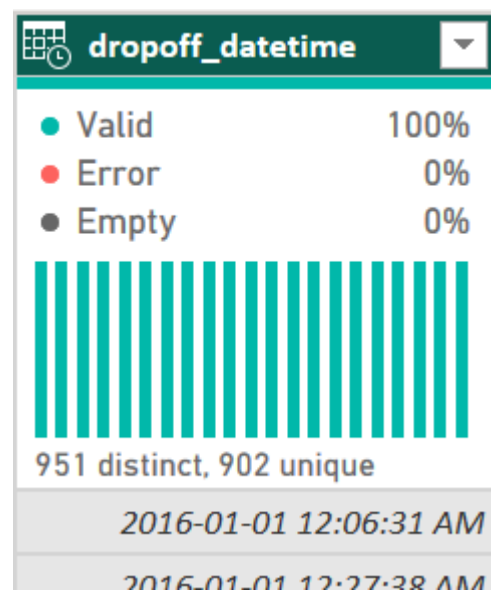
Original



Transformation



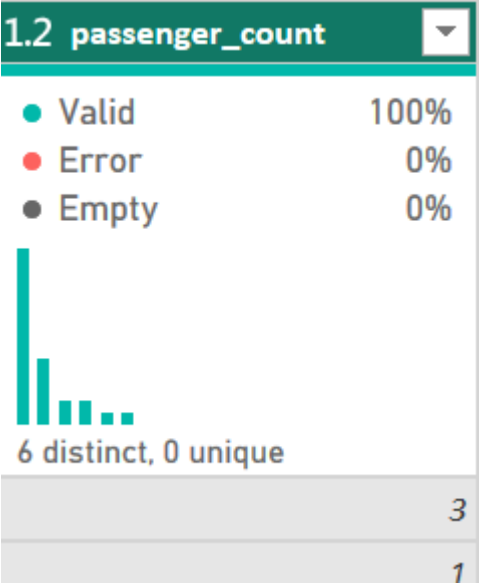
Final



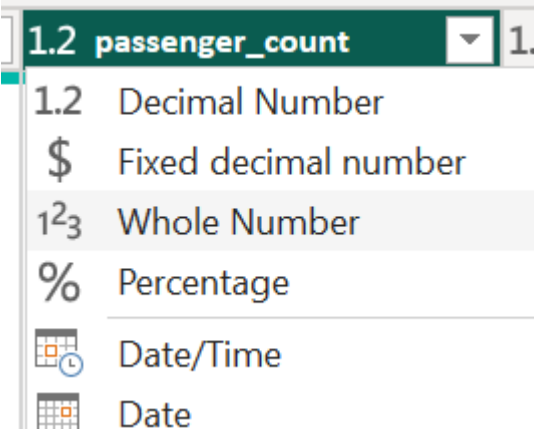
Passenger Count

This needs to be whole numbers

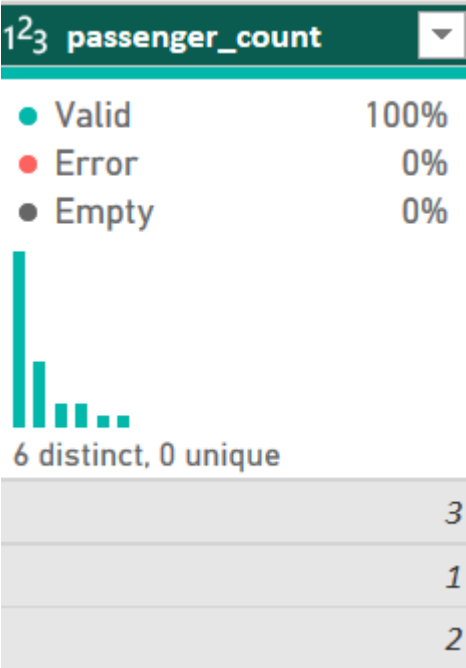
Original



Transformation



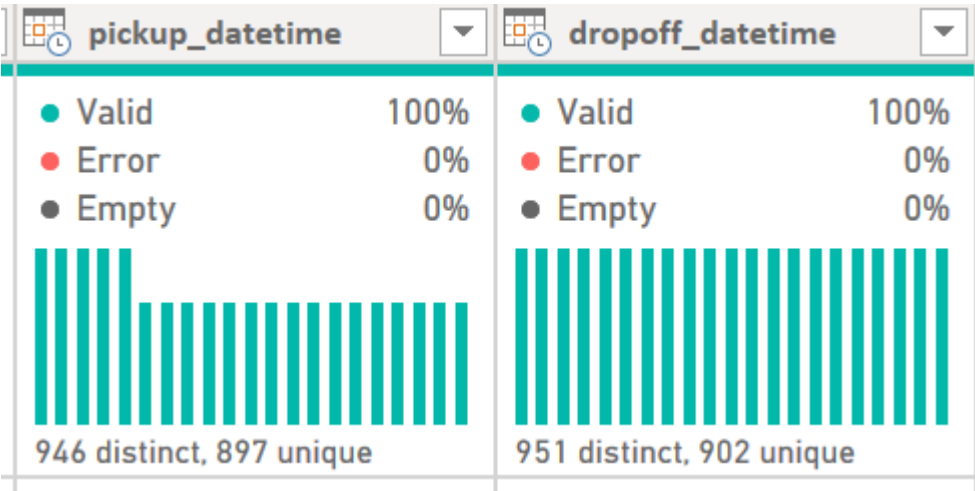
Final



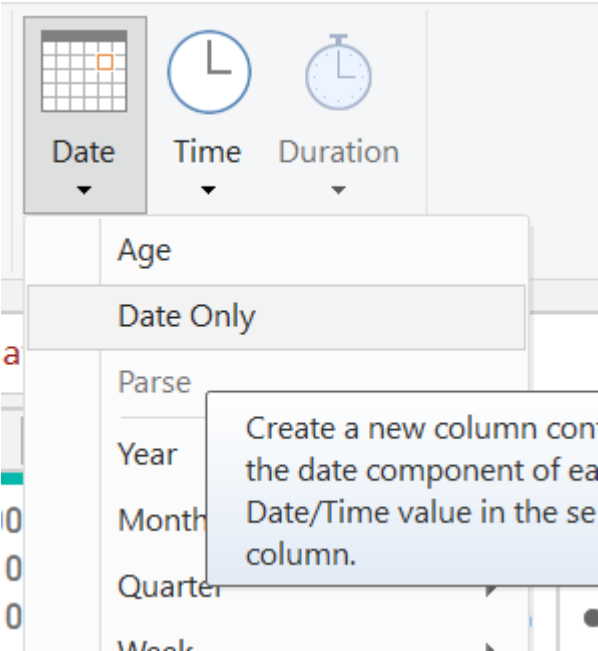
Splitting datetime columns

Date Only Columns

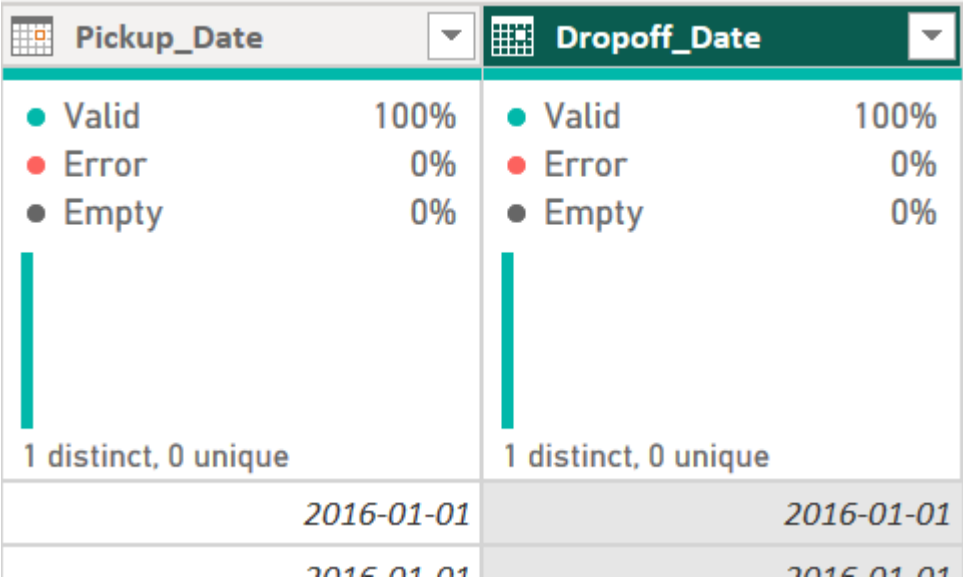
Original



Transformation

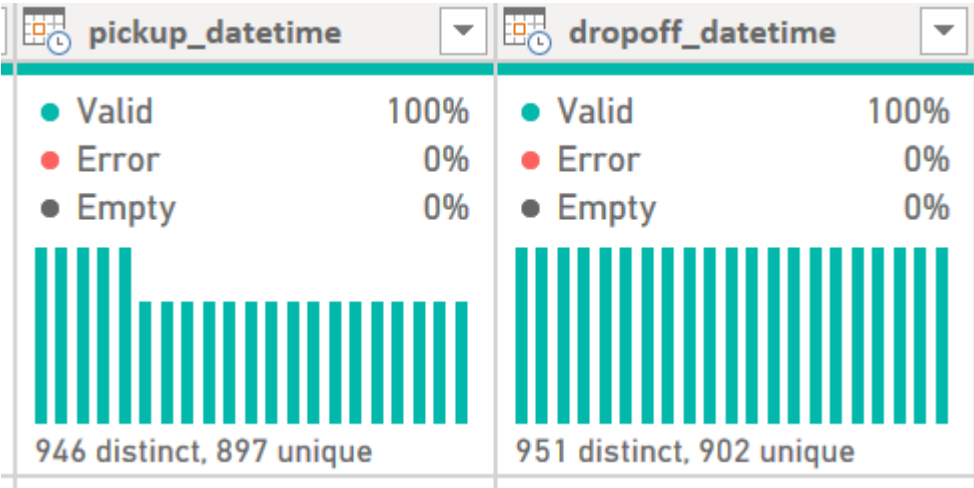


Final

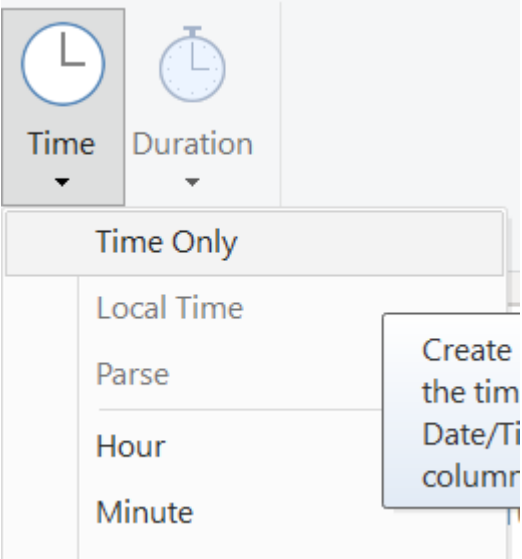


Time Only Columns

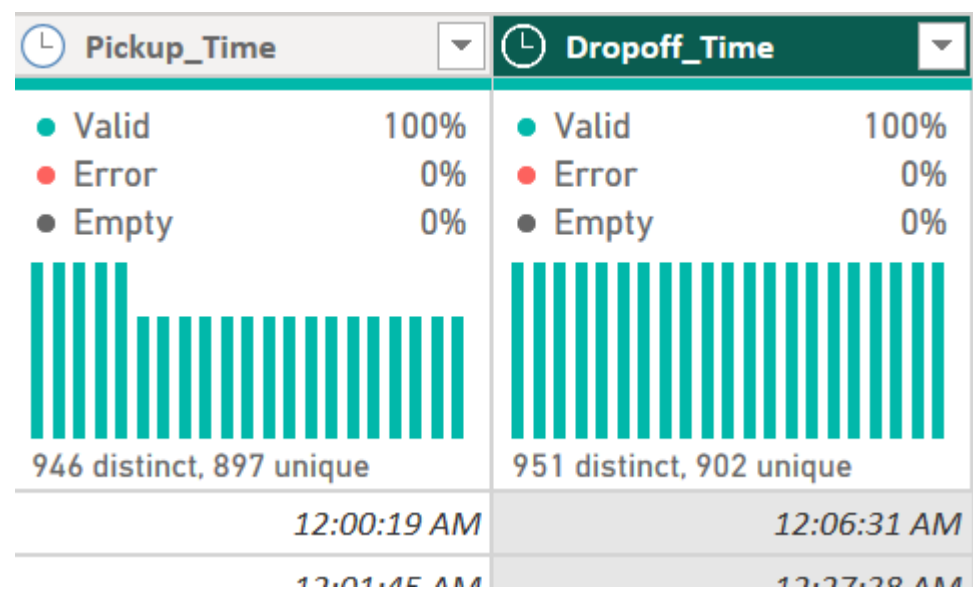
Original



Transformation



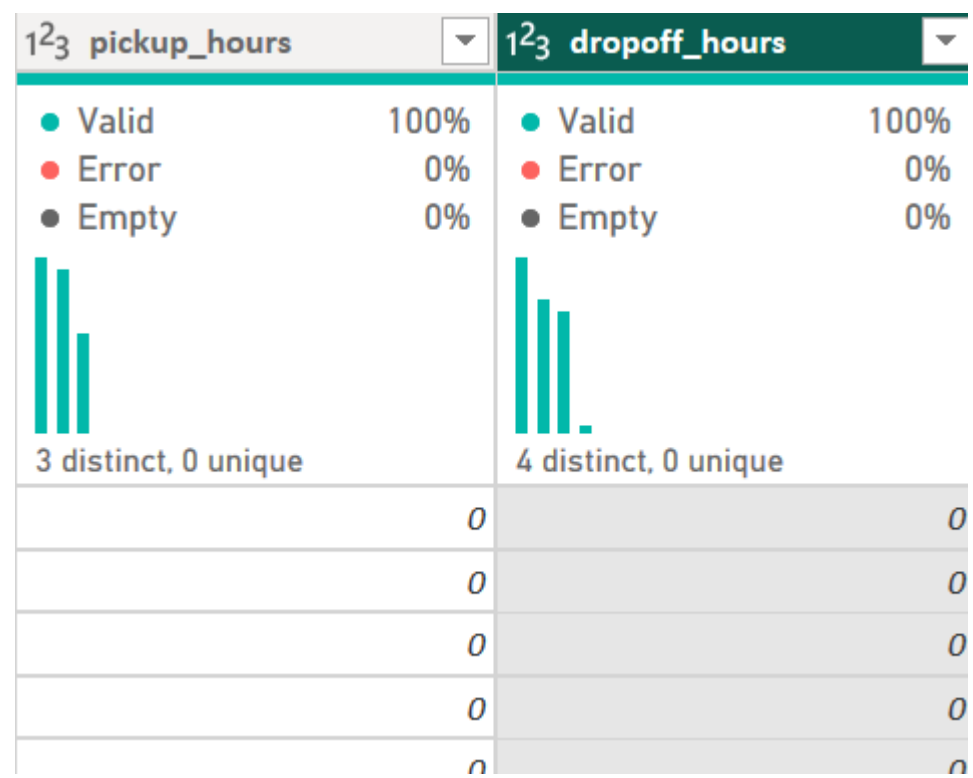
Final



Creating Conditional Column (Peak Hours)

Creating a new column called pickup and dropoff time Time Category labelling peak hours and off peak hours

Original



Transformation

Custom Column

Add a column that is computed from the other columns.

New column name

Pickup Time Category

Custom column formula ⓘ

```
= if [pickup_hours] >= 7 and [pickup_hours] < 10 then
  "Morning Peak"
else if [pickup_hours] >= 16 and [pickup_hours] < 19 then
  "Evening Peak"
else if [pickup_hours] >= 10 and [pickup_hours] < 16 then
  "Midday"
else "Night"
```

Available columns

id

vendor_id

pickup_datetime

dropoff_datetime

passenger_count

trip_distance

payment_type

trip_duration

<< Insert

Learn about Power Query formulas

✔ No syntax errors have been detected.

OK

Cancel

Custom Column

Add a column that is computed from the other columns.

New column name

Dropoff Time Category

Custom column formula ⓘ

```
= if [dropoff_hours] >= 7 and [dropoff_hours] < 10 then
  "Morning Peak"
else if [dropoff_hours] >= 16 and [dropoff_hours] < 19
then "Evening Peak"
else if [dropoff_hours] >= 10 and [dropoff_hours] < 16
then "Midday"
else "Night"
```

Available columns

id

vendor_id

pickup_datetime

dropoff_datetime

passenger_count

trip_distance

payment_type

trip_duration

<< Insert

Learn about Power Query formulas

✔ No syntax errors have been detected.

OK

Cancel

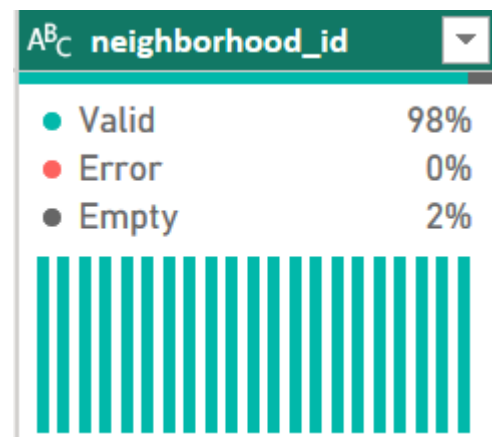
Final

AB_C Pickup Time Category	AB_C Dropoff Time Category
Valid100% Error0% Empty0% 1 distinct, 0 unique	Valid100% Error0% Empty0% 1 distinct, 0 unique
Night	Night
Night	Night
Night	Night
Night	Night
Night	Night
Night	Night

Handling Null Values

DIM_Dropoff_hoods Table

Original We can see from the image that we indeed have nulls that need to be handled



Transformation

REMOVING EMPTY

Final No more NULLS



Original

Transformation

17 / 42

Sort Ascending

Sort Descending

Clear Sort

Clear Filter

Remove Empty

Number Filters

Search

☒ (Select All)

☒ 0

☒ 0.01

☒ 0.09

☒ 0.1

☒ 0.2

1-3 payment_type

Valid 100%

Error 0%

Empty 0%

Equals...

Does Not Equal...

Greater Than...

Greater Than Or Equal To...

Less Than...

Less Than Or Equal To...

Between...

Final

We can now see that there is no zero value

 Sort Ascending

 Sort Descending

Clear Sort

 Clear Filter

Remove Empty

Number Filters

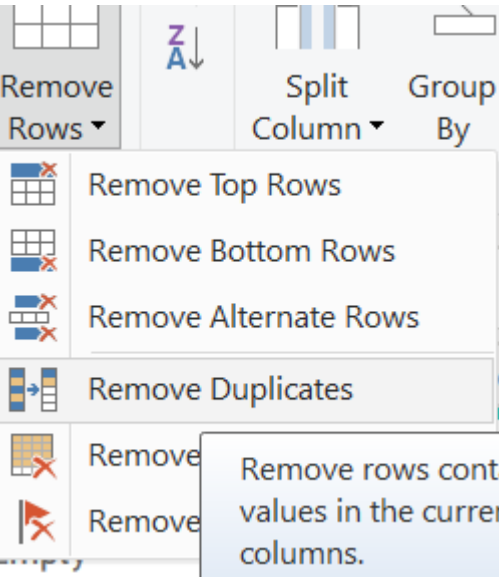
Search

- ☒ (Select All)
- ☒ 0.01
- ☒ 0.09
- ☒ 0.1
- ☒ 0.2
- ☒ 0.24
- ☒ 0.28
- ☒ 0.29
- ☒ 0.3

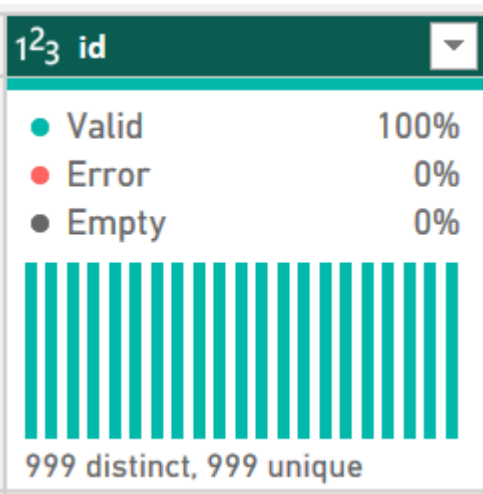
Removing Duplicates

The Dataset was too large to load accurate metrics but we removed duplicated from the ID section

Transformation



Final



Creating Custom Column(SPEED)

Speed column created using distance / duration /3600seconds to convert it into hours as the time in seconds is a little unseemly.

Transformation

Custom Column

Add a column that is computed from the other columns.

New column name

speed_mph

Custom column formula ⓘ

= ([trip_distance] / ([trip_duration]/3600))

Available columns

- id
- vendor_id
- pickup_datetime
- dropoff_datetime
- passenger_count
- trip_distance
- payment_type
- trip_duration

<< Insert

[Learn about Power Query formulas](#)

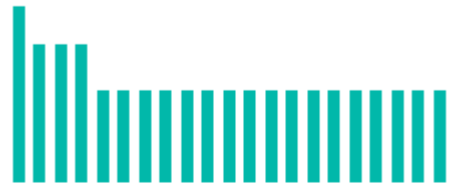
✓ No syntax errors have been detected.

OK

Cancel

\$ speed_mph

- Valid 100%
- Error 0%
- Empty 0%



965 distinct, 936 unique

12.77
31.76
15.85
30.17
9.70
18.50
6.72
22.99
18.04

Final

Merge Queries (Pickup Neighborhood Names)

Transformation



Merge

Select a table and matching columns to create a merged table.

FACT_trips



	trip_distance	payment_type	trip_duration	pickup_neighborhood	dropoff_neighborhood	Pickup_Date
3	1.32	2	372	AH	C	2016-01-
1	13.7	1	1553	Z	S	2016-01-
2	5.3	2	1204	D	AL	2016-01-
1	7.19	2	858	AT	J	2016-01-
0	0.0	1	1076	AG	...	2016-01-

DIM_pickup_hoods



neighborhood_id	latitude	longitude
AH	40.80434902	-73.96171613
Z	40.71582831	-73.95429804
D	40.75017915	-73.99255731
AT	40.72966953	-73.98169332
AG	40.74984348	-74.00345827

Join Kind

Left Outer (all from first, matching from second)

☐ Use fuzzy matching to perform the merge

▸ Fuzzy matching options

✓ The selection matches 999 of 999 rows from the first table.

OK

Cancel

Search Columns to Expand

A

Z

☒ Expand ☐ Aggregate

☒ (Select All Columns)

☒ neighborhood_id

☐ latitude

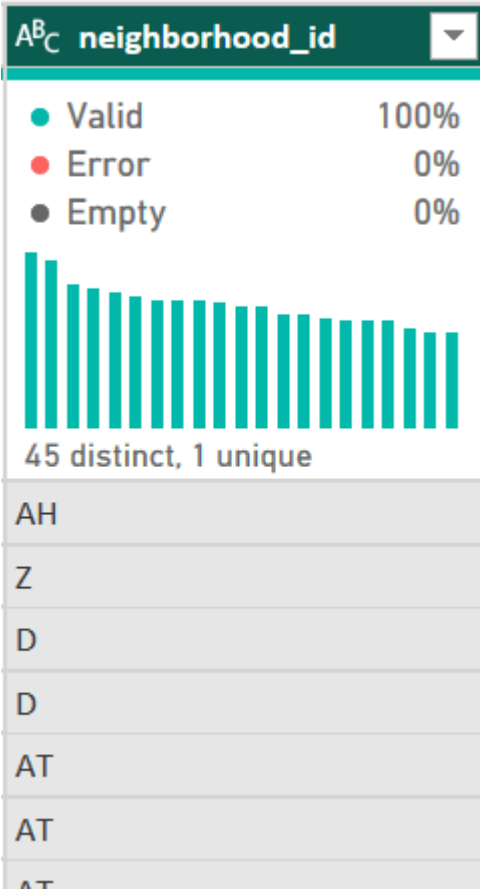
☐ longitude

☐ Use original column name as prefix

OK

Cancel

Final



Merge Queries (Dropoff Neighborhood Names)

Transformation

Merge

Select a table and matching columns to create a merged table.

FACT_trips

stance	payment_type	trip_duration	pickup_neighborhood	dropoff_neighborhood	Pickup_Date	Dropo
1.32	2	372	AH	C	2016-01-01	
13.7	1	1553	Z	S	2016-01-01	
5.3	2	1204	D	AL	2016-01-01	
0.69	1	316	D	D	2016-01-01	

DIM_dropoff_hoods

neighborhood_id	latitude	longitude
C	40.78378035	-73.95314487
S	40.83679153	-73.93438102
AL	40.81844494	-73.94804587
J	40.79957292	-73.94042746
AV	40.77627046	-73.98232156

Join Kind

Left Outer (all from first, matching from second)

☐ Use fuzzy matching to perform the merge

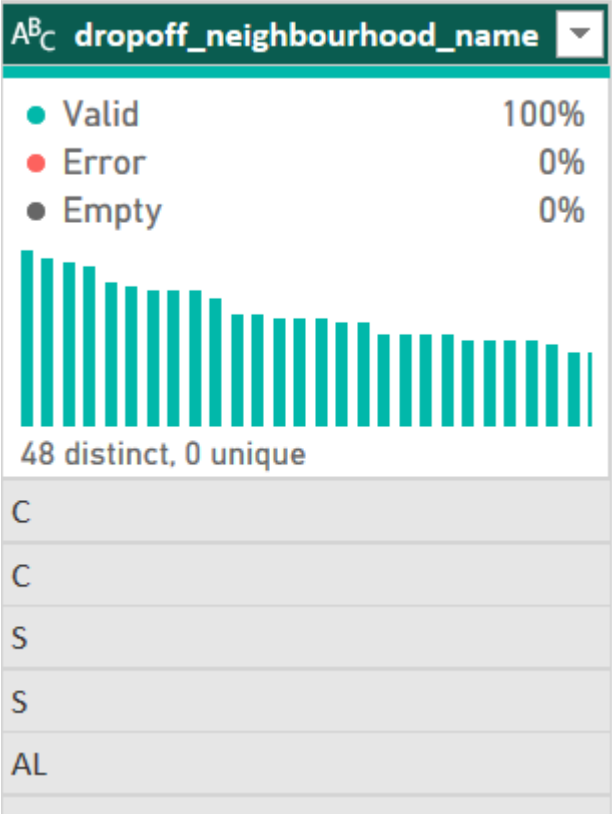
▸ Fuzzy matching options

✓ The selection matches 999 of 999 rows from the first table.

OK

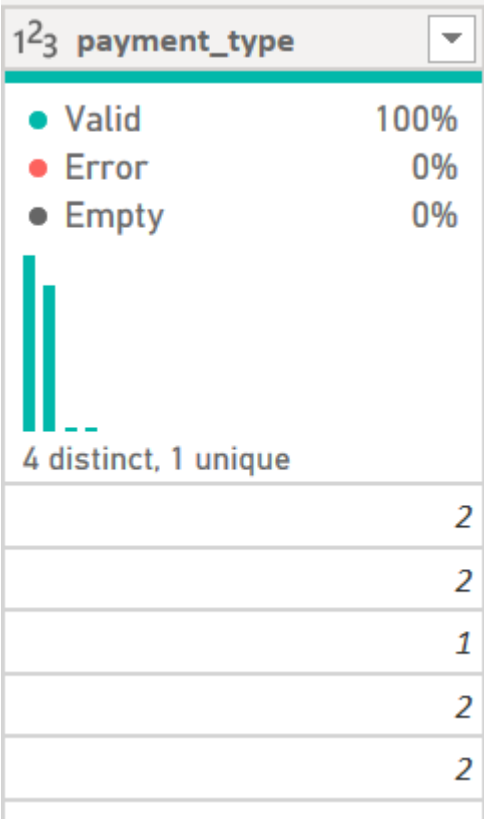
Cancel

Final



Replace Values (Decoding Payment Types)

Original



Transformation

Replace Values

Replace one value with another in the selected columns.

Value To Find

Replace With

▷ Advanced options

Replace Values

Replace one value with another in the selected columns.

Value To Find

Replace With

▷ Advanced options

Replace Values

Replace one value with another in the selected columns.

Value To Find

Replace With

▷ Advanced options

Replace Values

Replace one value with another in the selected columns.

Value To Find

4

Replace With

disputed

▸ Advanced options

Final

Text Filters

Search

☒ (Select All)

☒ cash

☒ credit card

☒ disputed

☒ no charge

OK

Cancel

Creating Dimensional Tables

Creating DIM_Date

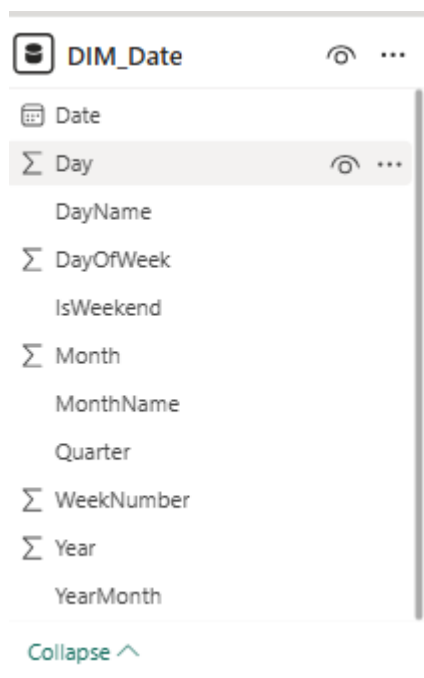
Transformation

```

1 DIM_Date =
2 VAR StartDate = MIN(FACT_trips[Pickup_Date])
3 VAR EndDate = MAX(FACT_trips[Pickup_Date])
4 RETURN
5
6 ADDCOLUMNS(
7     CALENDAR(StartDate, EndDate),
8     "Year", YEAR([Date]),
9     "Month", MONTH([Date]),
10    "MonthName", FORMAT([Date], "MMMM"),
11    "Day", DAY([Date]),
12    "DayOfWeek", WEEKDAY([Date]),
13    "DayName", FORMAT([Date], "DDDD"),
14    "WeekNumber", WEEKNUM([Date]),
15    "IsWeekend", IF(WEEKDAY([Date]) IN {1,7}, TRUE, FALSE),
16    "Quarter", "Q" & QUARTER([Date]),
17    "YearMonth", FORMAT([Date], "YYYY-MM")
18 )

```

Final



Create DimTime (for hour-level analysis)

Transformation

```

1 DIM_Time =
2 GENERATESERIES(0, 23, 1)

```

```

1 TimeCategory =
2 SWITCH( TRUE(),
3     DIM_Time[hour] >= 7 && DIM_Time[hour] < 10, "Morning Peak",
4     DIM_Time[hour] >= 10 && DIM_Time[hour] < 16, "Midday",
5     DIM_Time[hour] >= 16 && DIM_Time[hour] < 19, "Evening Peak",
6     "Night"
7 )
-

```

Final

DIM_Time	
Σ hour	
TimeCategory	

Collapse ^

Create DimPayment (to decode payment types)

Transformation

```

DIM_Payment =
DATATABLE (
    "PaymentCode", INTEGER,
    "PaymentMethod", STRING,
    {
        { 1, "Credit Card" },
        { 2, "Cash" },
        { 3, "No Charge" },
        { 4, "Dispute" }
    }
)

```

Final

 DIM_Payment

 ...

 PaymentCode

 PaymentMethod

Collapse ^

Creating Relationship Models

- Dim_date to FACT

New relationship

Select tables and columns that are related.

From table

DIM_Date

Date	Day	DayName	DayOfWeek	IsWeekend	Month	MonthName
01/01/2016 1:00:05 AM	1	Friday	6	false	1	January

To table

FACT_trips

Pickup_Date	pickup_datetime	pickup_hours	pickup_neighborhood	pickup_neighborhood	Pickup_Time	trip_distance
January-01-16	01/01/2016 1:00:05 AM	1	K	K	1:00:05 AM	3.3
January-01-16	01/01/2016 1:03:14 AM	1	E	E	1:03:14 AM	6.7
January-01-16	01/01/2016 1:04:04 AM	1	Y	Y	1:04:04 AM	0.68

Cardinality

One to many (1:*)

Cross-filter direction

Single

☒ Make this relationship active

☐ Apply security filter in both directions

☐ Assume referential integrity

Save

Cancel

- Dim_hour to FACT

New relationship



Select tables and columns that are related.

From table

DIM_Time

▼

hour	TimeCategory
0	Night
1	Night
2	Night

To table

FACT_trips

▼

ent_type	Pickup Time ...	Pickup_Date	pickup_dateti...	pickup_hours	pickup_neigh...	pickup_neigh...
t card	Night	January-01-16	01/01/2016 1:...	1	K	K
t card	Night	January-01-16	01/01/2016 1:...	1	E	E
t card	Night	January-01-16	01/01/2016 1:...	1	Y	Y

Cardinality

One to many (1:*)

▼

Cross-filter direction

Single

▼

☒ Make this relationship active

☐ Apply security filter in both directions

☐ Assume referential integrity

Save

Cancel

- Dim_Payment to FACT

New relationship

×

Select tables and columns that are related.

From table

DIM_Payment

PaymentCode	PaymentMet...
1	Credit Card
2	Cash
3	No Charge

To table

FACT_trips

f_Time	id	passenger_co...	payment_type	Pickup Time ...	Pickup_Date	pickup_dateti...
AM	375	1	credit card	Night	January-01-16	01/01/2016 1:...
AM	390	1	credit card	Night	January-01-16	01/01/2016 1:...
AM	395	1	credit card	Night	January-01-16	01/01/2016 1:...

Cardinality

One to many (1:*)

Cross-filter direction

Single

☒ Make this relationship active

☐ Assume referential integrity

☐ Apply security filter in both directions

Save

Cancel

- DIM Dropoffs to FACT

New relationship

✕

Select tables and columns that are related.

From table

DIM_dropoff_hoods

▼

latitude	longitude	neighborhoo...
40.7837803542	-73.95314486...	C
40.836791527	-73.93438102...	S
40.8184449425	-73.94804587...	AL

To table

FACT_trips

▼

Dropoff Time ...	Dropoff_Date	dropoff_datet...	dropoff_hours	dropoff_neig...	dropoff_neig...	Dropoff_Ti
Night	January-01-16	01/01/2016 1:...	1	H	H	1:16:56 /
Night	January-01-16	01/01/2016 1:...	1	M	M	1:24:37 /
Night	January-01-16	01/01/2016 1:...	1	AT	AT	1:08:20 /

Cardinality

One to many (1:*)

▼

Cross-filter direction

Single

▼

- ☒ Make this relationship active
- ☐ Assume referential integrity
- ☐ Apply security filter in both directions

Save

Cancel

- DIM Pickup to FACT

New relationship

×

Select tables and columns that are related.

From table

DIM_pickup_hoods

latitude	longitude	neighborhoo...
40.8043490199	-73.96171613...	AH
40.7158283067	-73.95429804...	Z
40.75017915	-73.99255731...	D

To table

FACT_trips

up Time ...	Pickup_Date	pickup_dateti...	pickup_hours	pickup_neigh...	pickup_neigh...	Pickup_Time
t	January-01-16	01/01/2016 1:...	1	K	K	1:00:05 AM
t	January-01-16	01/01/2016 1:...	1	E	E	1:03:14 AM
t	January-01-16	01/01/2016 1:...	1	Y	Y	1:04:04 AM

Cardinality

One to many (1:*)

Cross-filter direction

Single

☒ Make this relationship active

☐ Apply security filter in both directions

☐ Assume referential integrity

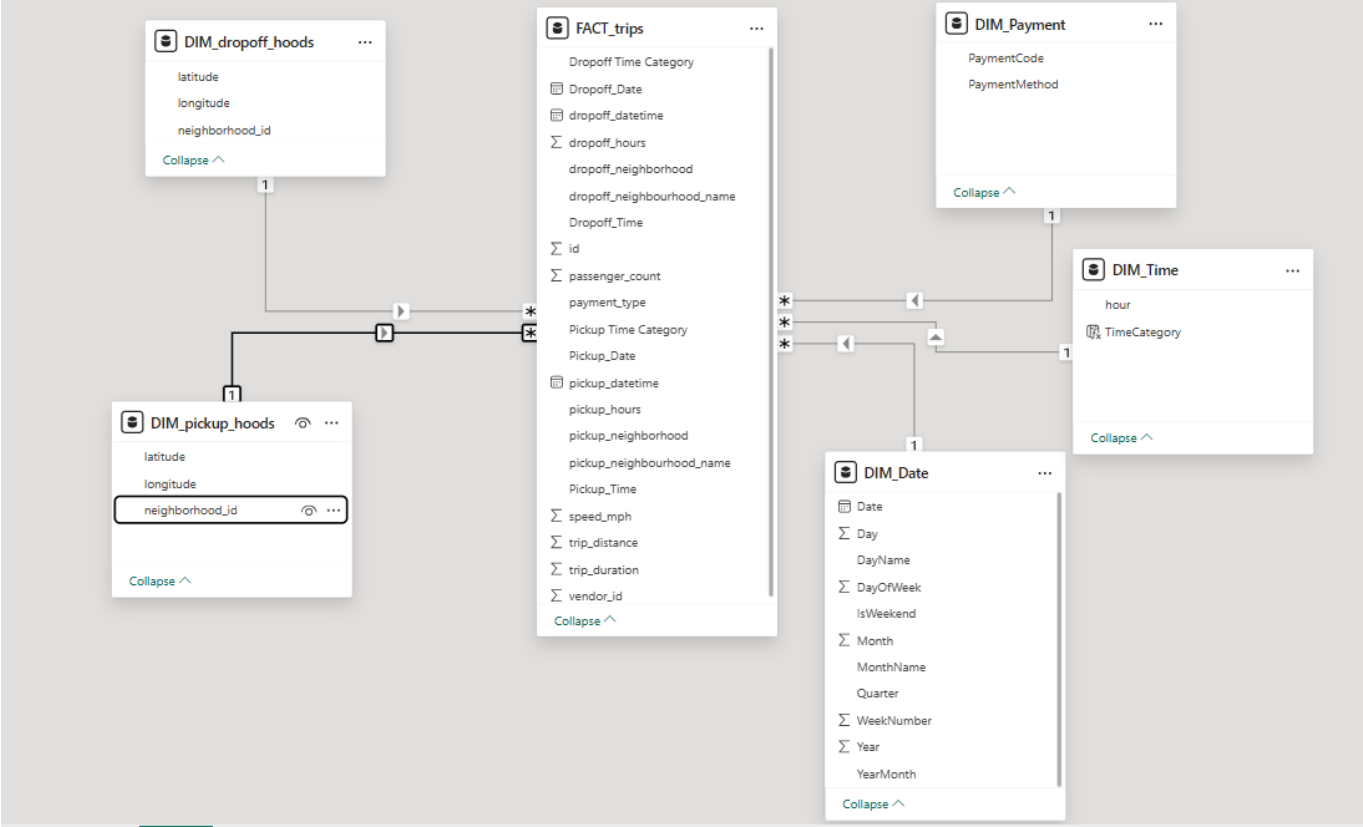
Desktop 2

Save

Cancel

STAR SCHEMA

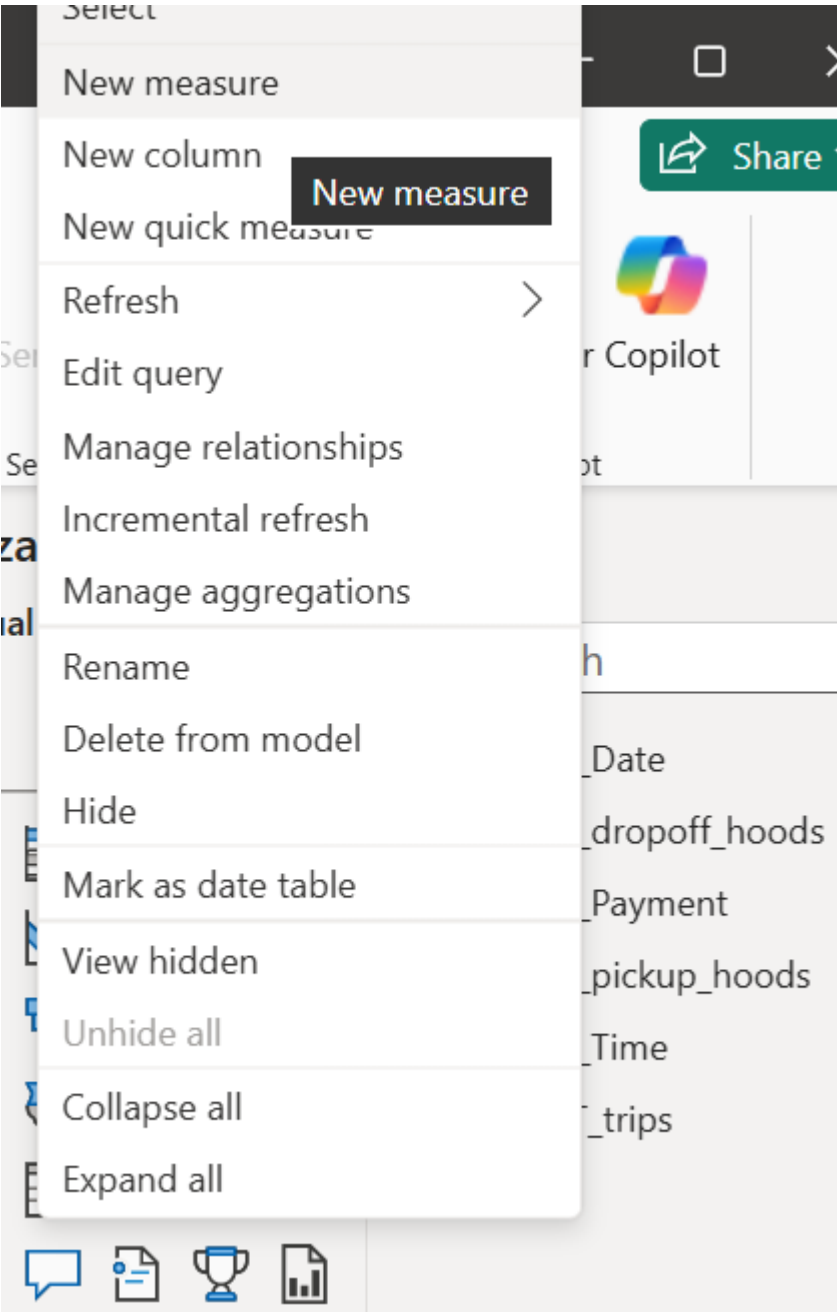
Dimension to FACT is always one to MANY!



DAX DEVELOPMENT:

New Measures

Total Trips



Transformation

Structure	Formatting
1	Total_Trips = COUNTROWS(FACT_trips)

Final

	Σ speed_mph
	☞ Total_Trips ...

Total Distance

Transformation

Structure	Formatting
	<pre>1 Total_Distance = SUM(FACT_trips[trip_distance])</pre>

Average Passengers

Transformation

Structure	Formatting
	<pre>1 Average_Passengers = AVERAGE(FACT_trips[passenger_count])</pre>

Year to Date YTD

Structure	Formatting
✓	<pre>1 Trips_YTD = 2 TOTALYTD(3 [Total_Trips], 4 DIM_Date[Date] 5)</pre>

Transformation

Month to Date MTD

Transformation

✓	<pre>1 Trips_MTD = 2 TOTALMTD(3 [Total_Trips], 4 DIM_Date[Date] 5)</pre>
-------------------------	--

Trips Year over Year Growth

Transformation

```

1 Trips_YoY_Growth =
2 VAR CurrentYearTrips = [Total_Trips]
3 VAR PreviousYearTrips =
4     CALCULATE(
5         [Total_Trips],
6         SAMEPERIODLASTYEAR(DIM_Date[Date]))
7     )
8 RETURN
9     DIVIDE(
10        CurrentYearTrips - PreviousYearTrips,
11        PreviousYearTrips,
12        0
13    )

```

Trips quarter over quarter Change

Transformation

```

1 Trips QoQ Change =
2 VAR CurrentQuarter = [Total_Trips]
3 VAR Previous_Quarter =
4     CALCULATE(
5         [Total_Trips],
5         PREVIOUSQUARTER(DIM_Date[Date]))
7     )
3 RETURN
3     CurrentQuarter - Previous_Quarter

```

Credit Card Percentage

Transformation

```

1 Credit Card % =
2 DIVIDE(
3     CALCULATE([Total_Trips], FACT_trips[payment_type] = 1),
4     [Total_Trips],
5     0
6 )

```

Cash percentage

Transformation

```
1 Cash % =  
2 DIVIDE(  
3     CALCULATE([Total_Trips], FACT_trips[payment_type] = 2),  
4     [Total_Trips],  
5     e  
6 )
```

Average Trip Duration (minutes)

Transformation

```
Avg_Duration_Minutes = AVERAGE(FACT_trips[trip_duration]) / 60
```

Peak Hour Trips

Transformation

```
1 Peak Hour Trips =  
2 CALCULATE([Total_Trips], DIM_Time[TimeCategory] = "Peak")
```

Speed Categories

Transformation

```
1 Speed Category =  
2 VAR AvgSpeed = AVERAGE(FACT_trips[speed_mph])  
3 RETURN  
4     SWITCH(  
5         TRUE(),  
6         AvgSpeed > 30, "Fast",  
7         AvgSpeed > 15, "Normal",  
8         "Slow"  
9     )
```

Part 2: Dashboard

Original

Transformation

Final

Total Trips

Transformation

Final

TEXT

Original

Transformation

Final

Total Trips

Transformation

Final

TEXT

Original

Transformation

Final

Total Trips

Transformation

Final

TEXT

Original

Transformation

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Total Trips

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